UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

ASSIGNMENT #5

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: *Velia Palma*
Due Date: Saturday April 04 by 12:00 noon

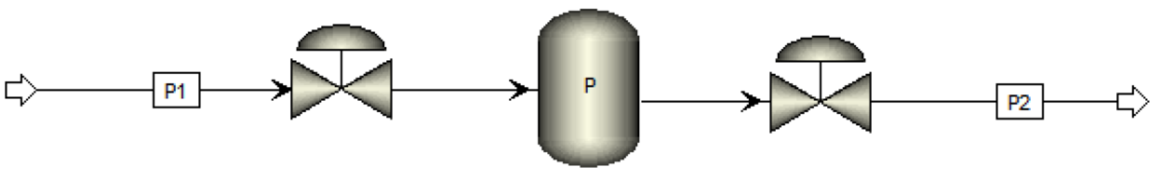
Number of points........................100
Your score.............................

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Scan your Assignment and create a PDF file
- PDF File Name: Assignment5_XY.PDF
  - X: First letter of your first name
  - Y: Your last name
  - Example: Assignment5_MCaracotsios.PDF
- Email your PDF file to: Daniel Christiansen dchris28@uic.edu and Yuechen Gao ygao54@uic.edu by the due date (Saturday April 04) and time (12:00 noon)

Note: **All problems are worth 100 points. You final score will be the average value**

Problem 5A: Consider the gas storage tank in the process flow diagram below. The volume of the tank is V and the temperature T is constant.



The molar gas flow rates are determined by the following equations:

$$\boxed{\begin{aligned} F_1 &= x_1 C_{V1}\sqrt{P_1 - P} \\ F_2 &= x_2 C_{V2}\sqrt{P - P_2} \end{aligned}}$$

where C_{V1} and C_{V2} are the valve coefficients and $\{x_1, x_2\}$ are the upstream and downstream valve openings. Assume the gas in the tank behaves as an ideal gas. Derive a transfer function model relating the tank pressure P to the inlet valve opening x_1. Develop expressions for the process gain K_P and lag time constant τ_P of the transfer function.

Molar Flow Rate

$$F = \left(\frac{PV}{RT}\right)\frac{1}{s}$$

Mass Balance

$$\frac{V}{RT}\frac{dP}{dT} = x_1 C_{V1}\sqrt{P_1 - P} - x_2 C_{V2}\sqrt{P - P_2} + \cancelto{0}{Generation}$$

At steady state

$$\frac{dP}{dT} = 0$$

$$0 = x_1 C_{V1}\sqrt{P_1 - P} - x_2 C_{V2}\sqrt{P - P_2}$$

$$y = P - P^{ss}$$

$$x = x_0 - x_1^{ss}$$

$$\frac{V}{RT}\frac{dP}{dT} = x_1 C_{V1}\sqrt{P_1 - P} - x_1^{ss} C_{V1}\sqrt{P_1 - P^{ss}} - x_2 C_{V2}\sqrt{P - P_2} + x_2 C_{V2}\sqrt{P^{ss} - P_2}$$

Linearization

$$F(x, y) = F(x_{ss}) + F'(x_{ss})(x - x_{ss}) + F'(y_{ss})(y - y_{ss})$$

$$F_1 = x_1^{ss} C_{V1}\sqrt{P_1 - P^{ss}} - \frac{\frac{1}{2} x_1^{ss} C_{V1}(P - P^{ss})}{\sqrt{P_1 - P^{ss}}} + C_{V1}\sqrt{P_1 - P^{ss}}(x_1 - x_1^{ss})$$

$$F_2 = x_2 C_{V2}\sqrt{P^{ss} - P_2} - \frac{\frac{1}{2} x_2^{ss} C_{V2}(P - P^{ss})}{\sqrt{P^{ss} - P_2}}$$

Therefore,

$$\frac{V}{RT}\frac{dP}{dT} = x_1^{ss} C_{V1}\sqrt{P_1 - P^{ss}} - \frac{\frac{1}{2} x_1^{ss} C_{V1}(P - P^{ss})}{\sqrt{P_1 - P^{ss}}} + C_{V1}\sqrt{P_1 - P^{ss}}(x_1 - x_1^{ss}) - x_1^{ss} C_{V1}\sqrt{P_1 - P^{ss}}$$
$$- x_2 C_{V2}\sqrt{P^{ss} - P_2} - \frac{\frac{1}{2} x_2^{ss} C_{V2}(P - P^{ss})}{\sqrt{P^{ss} - P_2}} + x_2 C_{V2}\sqrt{P^{ss} - P_2}$$

$$\frac{V}{RT}\frac{dP}{dT} = -\frac{\frac{1}{2} x_1^{ss} C_{V1}(P - P^{ss})}{\sqrt{P_1 - P^{ss}}} + C_{V1}\sqrt{P_1 - P^{ss}}(x_1 - x_1^{ss}) - \frac{\frac{1}{2} x_2^{ss} C_{V2}(P - P^{ss})}{\sqrt{P^{ss} - P_2}}$$

$$\frac{V}{RT}\frac{dP}{dT} = C_{V1}\sqrt{P_1 - P^{ss}}(x) - \frac{\frac{1}{2} x_1^{ss} C_{V1}(y)}{\sqrt{P_1 - P^{ss}}} - \frac{\frac{1}{2} x_2^{ss} C_{V2}(y)}{\sqrt{P^{ss} - P_2}}$$

$$\frac{V}{RT}\frac{dP}{dT} = C_{V1}\sqrt{P_1 - P^{ss}}(x) - \left(\frac{\frac{1}{2} x_1^{ss} C_{V1}}{\sqrt{P_1 - P^{ss}}} - \frac{\frac{1}{2} x_2^{ss} C_{V2}}{\sqrt{P^{ss} - P_2}}\right)(y)$$

First order Model

$$G(p) = \frac{y}{x} = \frac{k_p}{\tau_p + 1}$$

$$\tau_p \frac{dy}{dt} + y = k_p x$$

$$\frac{V}{RT}\frac{dy}{dt} + \left(\frac{x_1^{ss} C_{V1}}{2\sqrt{P_1 - P^{ss}}} - \frac{x_2^{ss} C_{V2}}{2\sqrt{P^{ss} - P_2}}\right)(y) = C_{V1}\sqrt{P_1 - P^{ss}}(x)$$

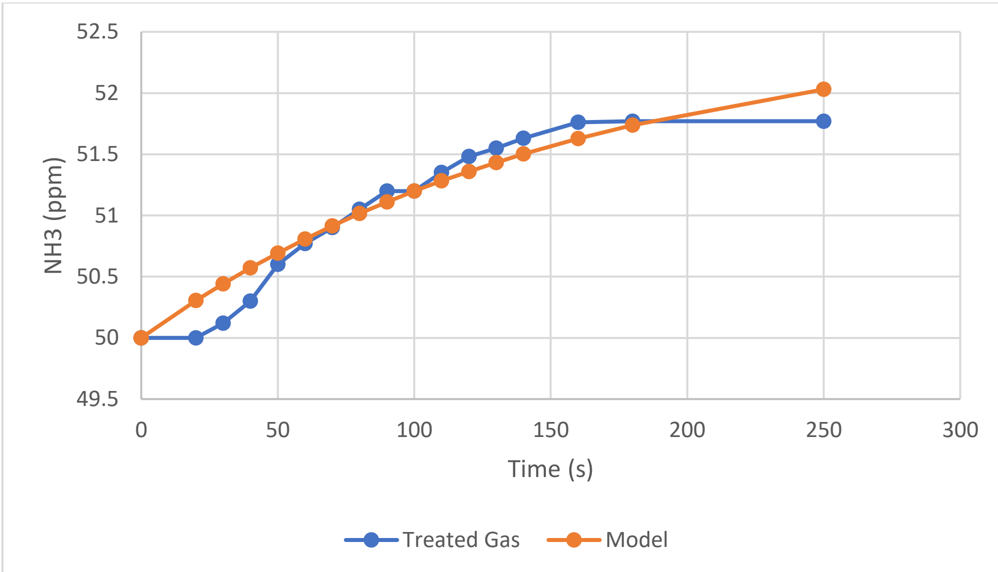
Divide by $\left(\frac{x_1^{ss} C_{V1}}{2\sqrt{P_1 - P^{ss}}} - \frac{x_2^{ss} C_{V2}}{2\sqrt{P^{ss} - P_2}}\right)$

$$\frac{V}{RT}\frac{dy}{dt} + y = \frac{C_{V1}\sqrt{P_1 - P^{ss}}}{\left(\frac{x_1^{ss} C_{V1}}{2\sqrt{P_1 - P^{ss}}} - \frac{x_2^{ss} C_{V2}}{2\sqrt{P^{ss} - P_2}}\right)} x$$

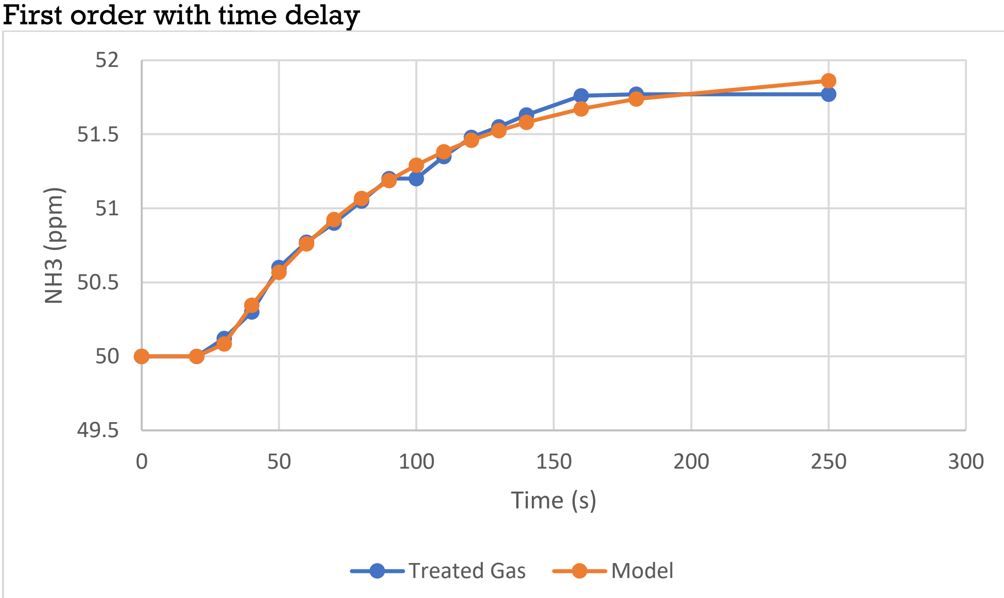
Problem 5B

Model discrimination study

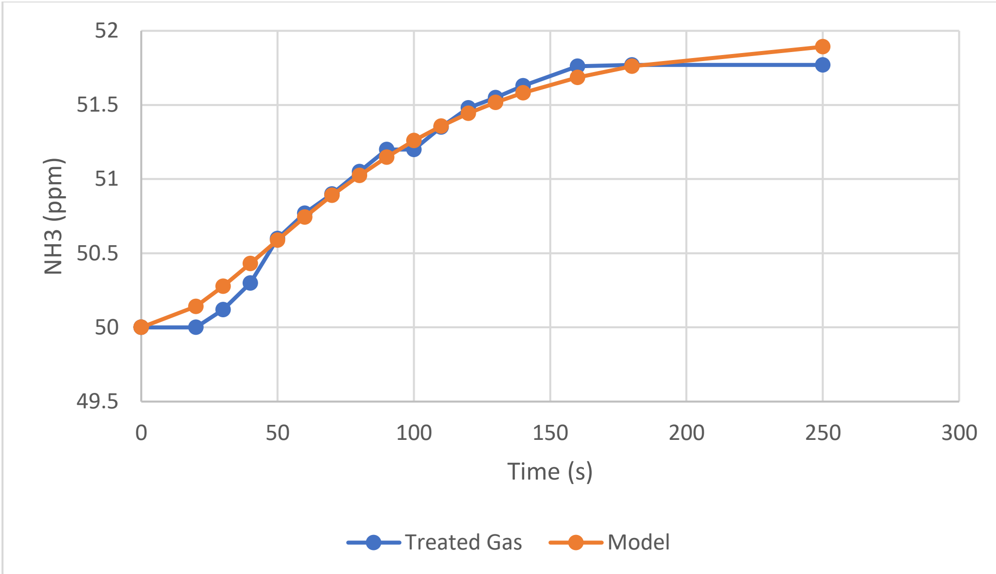
1. First order



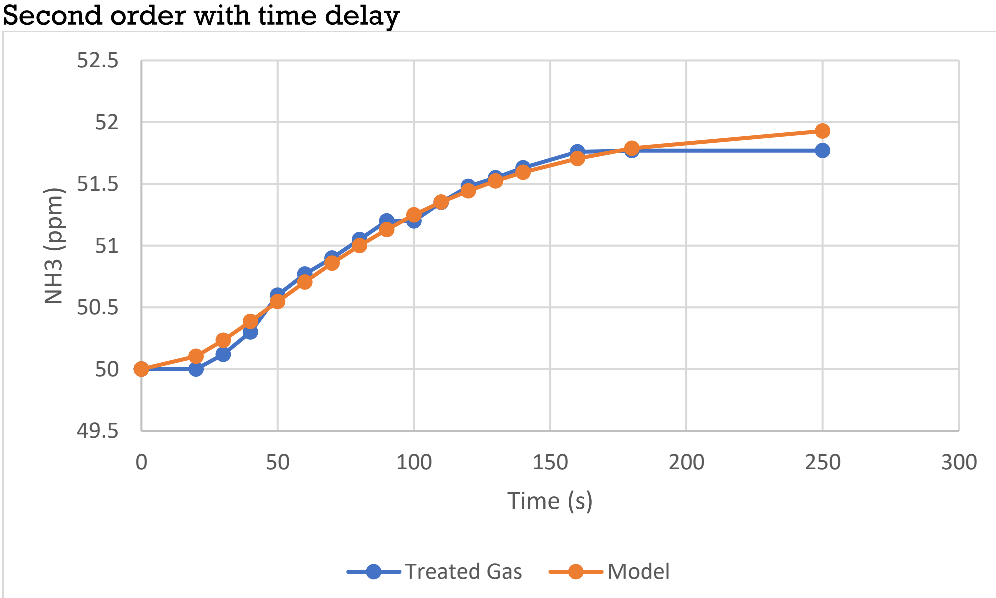
2. First order with time delay



3. Second order



4. Second order with time delay



Summary of results

Model	Kp	τp1	τp2	θ	RSQ	AIC
1	-0.0511	157.4			0.425	-59.9
2	-0.0385	65.6		27.24	0.0355	-102.7
3	-0.0385	45.2	31.7163		0.0973	-82.89
4	-0.0397	58.7	14.94	3.54	0.0810	-82.25